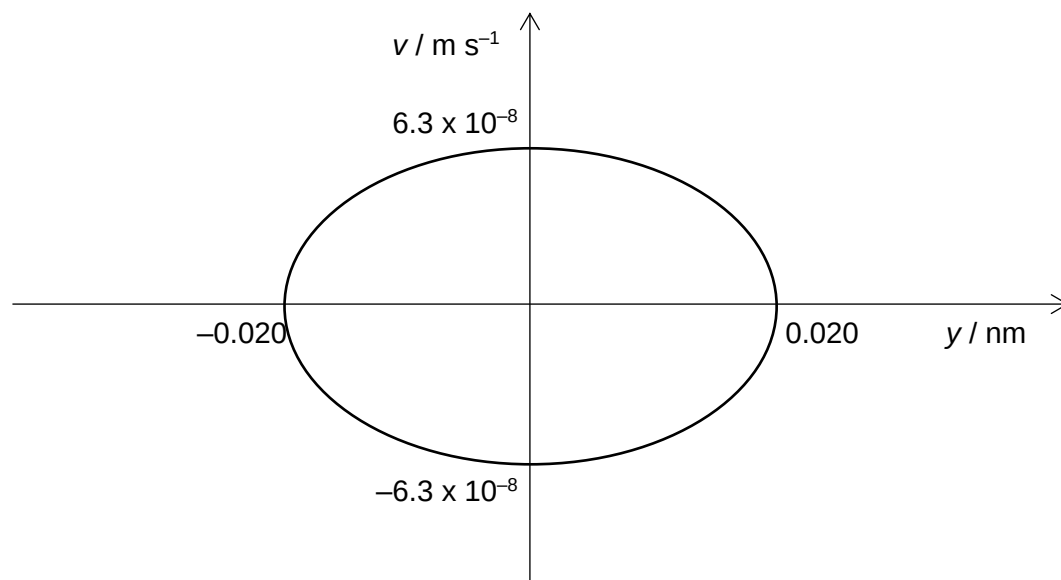


- 7 (a) progressive: energy transferred along direction of propagation of wave B1
 longitudinal: particles along wave oscillate along axis parallel to direction of propagation of waves B1
- (b) (i) amplitude = 2.0×10^{-11} m A1
 frequency = 500 Hz A1
- (ii) maximum speed
 $= \omega x_0 = (2\pi \times 500) \times 2 \times 10^{-11}$ M1
 $= 6.3 \times 10^{-8} \text{ m s}^{-1}$ A0
- (iii) maximum kinetic energy = $\frac{1}{2} m v_0^2$
 $\frac{1}{2} \times m \times (6.3 \times 10^{-8})^2 = 2.4 \times 10^{-19}$ C1
 $m = 1.2 \times 10^{-4} \text{ kg}$ A1
- (iv) ellipse / circle M1
 with correct axes labels A1



- (c) (i) energy transferred from beating wings to the ear / eardrum vibrates at same frequency as beating wing B1
 frequency within audible range of person (20 Hz – 20kHz) B1
- (ii) intensity $\propto \frac{1}{\text{distance}^2}$ C1
 $\frac{1.6I}{I} = \left(\frac{2.0}{2.0-x} \right)^2$ where I is original intensity C1
 $x = 0.42 \text{ m}$ A1

- (iii) air molecule travels $4 \times 5.0 \times 10^{-9}$ ($= 2.0 \times 10^{-8}$ m) each cycle C1
 total distance covered in 1800 beats $= 1800 \times 2.0 \times 10^{-8} = 3.6 \times 10^{-5}$ m A1
- (iv) device generate sound at same frequency as buzzing sound B1
 of equal / similar amplitude B1
 and 180° out-of-phase / anti-phase B1
 leads to destructive interference of buzzing sound and the generated sound B1
 that cancels the buzzing sound

